The problem statement:

> In a sci-fi film, Alex is a time traveler who decides to go back in time to witness a famous historical battle that took place 100 years ago, which lasted for 10 days. He arrives three days before the battle starts. However, after spending six days in the past, he jumps forward in time by 50 years and stays there for 20 days. Then, he travels back to witness the end of the battle. How many days does Alex spend in the past before he sees the end of the battle?

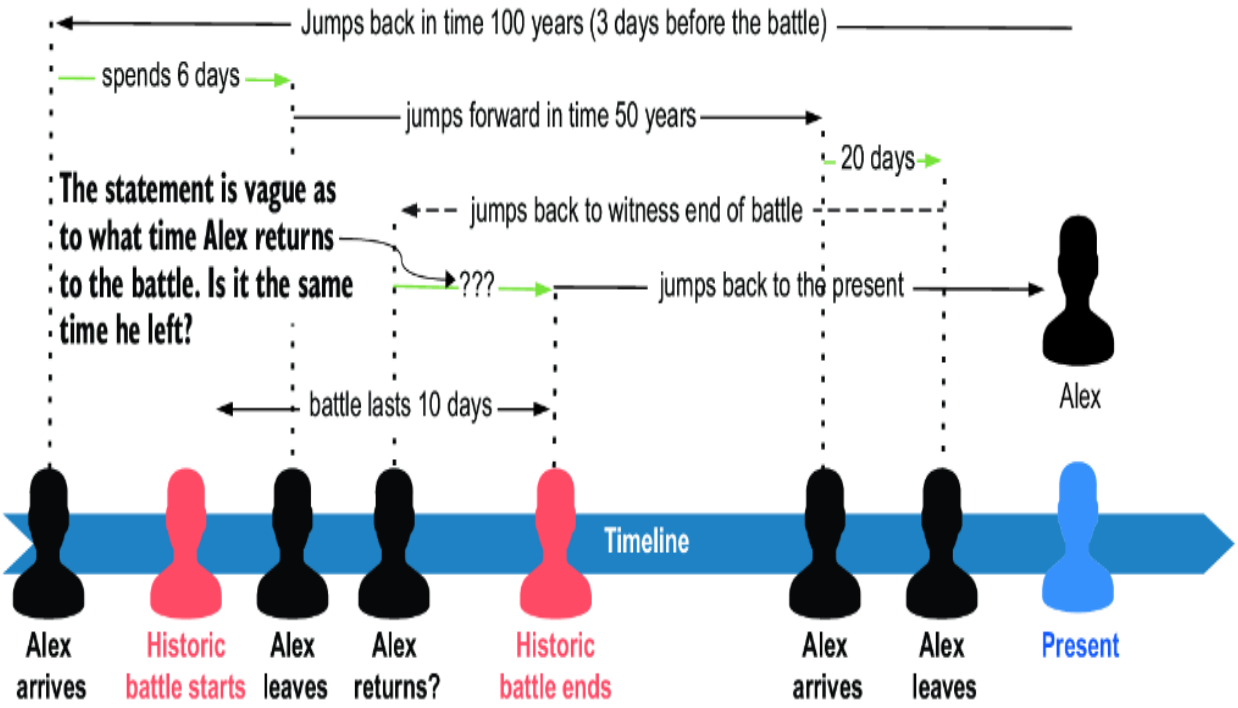



**Figure 10.3 The complexity of the time travel problems we intend to solve using LLMs with reasoning and planning**

Time travel problems are thought exercises that can be deceptively difficult to solve. The example in figure 10.3 is complicated to solve for an LLM, but the part it gets wrong may surprise you. The next section will use reasoning in prompts to solve these unique problems.

## 10.2.1 Chain of thought prompting

*Chain of thought* (CoT)prompting is a prompt engineering technique that employs the one-shot or few-shot examples that describe the reasoning and